

Climate Policy Uncertainty, Carbon Prices, and Volatility Risk in Clean Energy Equities: Evidence from the EU ETS

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Abstract

Europe’s energy transition hinges on sustained investment in clean energy, but investors still face an uncomfortable question: how risky are these assets when climate policy signals turn noisy? This paper examines how climate policy uncertainty (CPU) and conditions in the EU carbon market jointly shape volatility and tail risk in clean energy equities. Our central idea is simple: the same bad return can have very different volatility consequences depending on the policy and carbon-price backdrop. In particular, we expect negative shocks to translate into much stronger and longer-lasting volatility when policy uncertainty is high *and* EU Allowance (EUA) prices are falling than when policy looks steady and carbon prices are supportive. To test this, we use a nonlinear stochastic volatility model in which a small, stability-constrained neural network learns state-dependent volatility dynamics from the data while preserving the linear model as a special case. The nonlinear transition takes *observable* lagged returns—rather than latent standardized shocks—as inputs, which avoids the identification complications of feeding latent quantities into a flexible function and enables clean Bayesian estimation. We estimate the full model in Stan using Hamiltonian Monte Carlo (NUTS), exploiting its automatic differentiation to sample the joint posterior over latent log-variances and all model parameters, including the neural-network weights. We benchmark performance using out-of-sample density forecasts, tail-risk backtests (VaR/ES), and a volatility-timing portfolio exercise that converts statistical improvements into economic magnitudes. Interpretable state-dependent impulse responses and marginal-effect surfaces make the mechanisms visible and quantify how much volatility amplification is tied to policy uncertainty and carbon-market regimes. The results speak to both carbon-market design (credibility and price stability) and practical risk management for transition-exposed portfolios.

Keywords: climate policy uncertainty; EU ETS; carbon price; clean energy equities; volatility; tail risk; stochastic volatility; Bayesian estimation; Stan; neural network.

1 Introduction and Motivation

Reaching the EU’s climate-neutrality target by 2050 will require steady, large-scale investment in clean energy. The European Green Deal, Fit for 55, and REPowerEU all push the transition forward, and the EU Emissions Trading System (EU ETS) remains the flagship market-based policy tool. Yet ambition alone does not mobilize capital. Investors also need a clear view of risk, because risk shapes financing costs and, ultimately, the pace of deployment. The International Energy Agency estimates that global clean energy investment needs to roughly double by 2030 to stay on a net-zero pathway ([International Energy Agency, 2024](#)). If we want that money to show up, we need to understand what makes clean energy assets risky—and when.

A natural candidate is climate policy uncertainty (CPU). Clean energy firms are exposed to uncertainty about the future path of carbon prices, the durability of subsidies and tax

credits, permitting and grid-connection timelines, and the speed of fossil-fuel phase-outs. These uncertainties can move both expected cash flows and discount rates. A growing literature links CPU to mean returns and investment (Treepongkaruna et al., 2023; Liu et al., 2021). What we know much less about is the risk side: how CPU affects volatility, tail risk, and the dynamics of risk over time.

This paper targets that gap. Our core hypothesis is that the volatility response to return shocks is *state dependent*. It depends not only on whether returns are negative or positive, but also on the prevailing policy environment and the carbon-market signal. A negative return during a period of high CPU and falling EUA prices—a combination that can be read as wavering policy commitment—should trigger a larger and more persistent volatility increase than the same shock in a calmer, policy-supportive regime. If this is right, standard linear volatility models miss the very episodes that matter most for downside risk.

Why should energy economists care? This is not just a finance question. It connects directly to core energy-economics debates about credible policy and the cost of capital. First, it speaks to the idea that predictable carbon pricing can lower financing costs for clean energy (Bento and Gianfrate, 2020; Wen et al., 2022). If policy uncertainty amplifies volatility *nonlinearly* in adverse states, then the penalty for policy inconsistency may be much larger than linear estimates imply. Second, it adds a distributional perspective to work on spillovers between carbon and energy markets (Byun and Cho, 2023; Chevallier, 2009). Third, it matters for practitioners and regulators who increasingly treat transition risk as a financial-stability concern.

Approach. We model clean energy equity volatility with a stochastic volatility (SV) framework in which the latent log-variance evolves as a nonlinear function of its own lag, *observable* lagged returns, CPU, carbon price dynamics, and financial conditions. The choice to use raw lagged returns rather than latent standardized shocks is deliberate: it eliminates the need to feed an unobserved quantity into the flexible nonlinear function, which would create identification difficulties in estimation. The nonlinear piece is learned by a deliberately small neural network, equipped with stability constraints so it cannot generate explosive volatility paths. Importantly, the model nests the standard linear SV specification: if the nonlinear component is irrelevant, the data can effectively switch it off. We estimate the full joint posterior—latent volatility path and all parameters, including the NN weights—in Stan using the No-U-Turn Sampler (NUTS). We benchmark against standard approaches (GARCH variants, linear SV, and regime-switching SV) using out-of-sample density forecasts, tail-risk backtests, and a volatility-timing portfolio exercise. To keep the model economically interpretable, we summarize results with state-dependent impulse responses, marginal-effect surfaces, and ablation checks.

Preview of contributions. The paper contributes in four ways. (1) It documents how CPU maps into *volatility dynamics*—not just mean outcomes—for clean energy equities, and shows that this mapping depends on carbon-market conditions. (2) It quantifies the economic relevance of this state dependence by focusing on tail risk and adverse regimes. (3) It shows that a parsimonious nonlinear SV model, estimated via full Bayesian inference in Stan, can improve density and tail-risk forecasting relative to standard benchmarks in this setting. (4) It draws

policy implications that link carbon-market credibility and price stability to financial risk in transition assets.

2 Institutional Background: EU ETS and the Clean Energy Policy Landscape

The EU ETS, launched in 2005, is the world’s largest carbon market and a central pillar of EU climate policy. Because our analysis conditions on carbon-price states, it is useful to recall the ETS’s institutional evolution and how it has shaped price dynamics.

Phase structure and reforms. The EU ETS has changed substantially over time. Phase 1 (2005–2007) was an early learning period. Phase 2 (2008–2012) experienced a large allowance surplus and, as a result, weak prices. Phase 3 (2013–2020) moved toward EU-wide caps and expanded auctioning. A major turning point was the Market Stability Reserve (MSR), which became operational in 2019 and began withdrawing surplus allowances, helping restore the carbon price signal. Phase 4 (2021–2030) aligns more closely with the European Green Deal and Fit for 55 reforms, including expanded coverage (e.g., maritime shipping) and the introduction of ETS 2 for buildings and road transport.

Implications for this study. These reforms create clear episodes in which carbon prices behave differently and policy debates become especially salient. We begin the main sample in 2013 (Phase 3 onward) to focus on the period in which the ETS is widely viewed as a more credible price signal. We use the MSR introduction in 2019 and the Fit for 55 legislative process starting in 2021 as anchors for subsample and event-style robustness. More broadly, the moments that matter most for our hypothesis are precisely those where reform discussions, implementation delays, or debates over cap stringency raise CPU—and where we expect the strongest state-dependent volatility effects.

Carbon prices and clean energy firms. Carbon prices matter for clean energy firms in two ways. Mechanically, higher EUA prices raise the cost of emissions-intensive production and can improve the relative competitiveness of clean technologies. Informationally, carbon prices also serve as a market-based signal of policy credibility. A falling carbon price—especially during high CPU—can simultaneously weaken expected profitability and raise doubts about the long-run policy path. This dual role makes the EU ETS a natural conditioning variable for understanding how policy uncertainty shows up in clean energy equity risk.

3 Transmission Channels: From Policy Uncertainty to Volatility

Before turning to the econometric framework, we outline the economic channels linking CPU to volatility, and why these channels are likely to interact with the carbon price state. Figure 1 summarizes the logic.

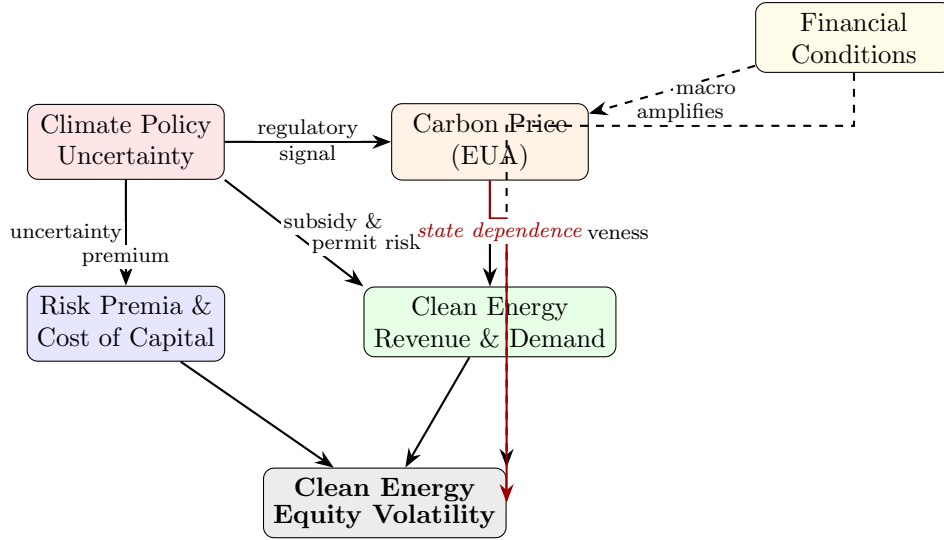


Figure 1: Transmission channels from climate policy uncertainty to clean energy equity volatility. Solid arrows: direct effects. Dashed arrows: moderating roles. The red path highlights the hypothesized state dependence: the carbon price state modulates how CPU shocks transmit into volatility.

Channel 1: Uncertainty premium. When climate policy becomes harder to predict, investors face a wider range of plausible futures for clean energy cash flows and discount rates. That widening should raise both perceived risk and the compensation investors demand (Bloom, 2009; Pástor et al., 2022). In practice, this shows up as higher volatility and fatter downside tails.

Channel 2: Carbon price signal. CPU can also destabilize the carbon-price signal itself. If policy debates create doubts about future cap tightness or the enforcement of reforms, EUA prices can weaken, and the investment case for clean energy becomes less secure (Koch et al., 2021; Hintermann, 2010). Importantly, falling carbon prices may be read as both a valuation hit and a credibility warning, which can strengthen the volatility response to bad news.

Channel 3: Financing conditions. Clean energy is capital intensive, often financed through project finance and long-dated cash flows. When credit spreads widen or risk-free rates rise, financing becomes more fragile and the same policy uncertainty shock can bite harder (Wen et al., 2022). This suggests an interaction between CPU and financial conditions.

Why state dependence? These channels likely reinforce one another in adverse states. High CPU plus falling EUA prices is not just “two negatives.” Together they can suggest a broader shift in the policy regime, triggering feedback loops: investors de-risk, financing terms worsen, projects are delayed, and valuations become more sensitive to new information. In contrast, high CPU paired with stable or rising carbon prices may be partly cushioned by the market’s signal that policy is still holding. Capturing this conditional amplification is exactly what linear models struggle to do.

4 Research Questions and Hypotheses

- RQ1.** Does climate policy uncertainty increase volatility and downside risk in clean energy equities, beyond what is explained by broad market uncertainty and energy price fluctuations?
- RQ2.** Is the volatility effect of CPU *state dependent* on carbon market conditions (EUA level/returns) and broader financial conditions?
- RQ3.** Does a nonlinear volatility model deliver meaningful improvements in forecasting tail risk relative to standard approaches, and are the gains concentrated in high-uncertainty regimes?
- RQ4.** What is the economic magnitude of state-dependent amplification, and what does it imply for the financial stability value of credible carbon pricing?

Hypotheses. Let y_t denote weekly clean energy equity returns, h_t latent log-variance, and x_t the conditioning set (CPU, EUA, financial conditions).

- H1. CPU raises volatility:** $\partial \mathbb{E}[h_{t+1} \mid \cdot] / \partial \text{CPU}_t > 0$, controlling for broad equity volatility and energy prices.
- H2. Carbon price state dependence:** The marginal effect of CPU is larger when EUA prices are low or declining: $\partial^2 \mathbb{E}[h_{t+1}] / \partial \text{CPU}_t \partial (-\Delta \log \text{EUA}_t) > 0$.
- H3. Financial conditions amplification:** $\partial^2 \mathbb{E}[h_{t+1}] / \partial \text{CPU}_t \partial \text{Spread}_t > 0$.
- H4. Asymmetric shock persistence:** Negative return shocks generate stronger, more persistent volatility increases under joint high CPU and adverse carbon/financial states.
- H5. Forecasting gains:** The nonlinear model improves out-of-sample density scores and tail-risk backtests, with gains concentrated in high-CPU episodes.

5 Related Literature

This study sits at the intersection of three literatures that are central to energy economics and energy finance.

Climate policy uncertainty and energy markets. CPU indices (PolicyUncertainty.com, 2026a; Figshare, 2024) build on the broader economic policy uncertainty framework (Baker et al., 2016) by focusing on climate-transition policy. Existing work links CPU to renewable investment (Liu et al., 2021), green finance (Wen et al., 2022), and the cross-section of stock returns (Treepongkaruna et al., 2023). The missing piece is the behavior of *risk*—how uncertainty changes volatility dynamics and tail outcomes, which are often the binding constraint for portfolio allocation and capital provision.

Carbon markets and financial transmission. A large literature studies EUA price drivers and the EU ETS’s interaction with the macroeconomy and energy markets (Hintermann, 2010; Creti et al., 2012; Chevallier, 2009). Related work examines how political events and reforms move carbon prices (Koch et al., 2021) and how risk spills over between carbon and energy assets (Byun and Cho, 2023). We take a complementary angle: rather than explaining carbon prices per se, we use carbon-market conditions as a *state variable* that governs how policy uncertainty feeds into equity volatility.

Volatility modeling in energy finance. GARCH-type models remain standard tools in applied energy finance (Bollerslev, 1986; Nelson, 1991; Glosten et al., 1993). Regime-switching models capture discrete shifts (Hamilton, 1989; So et al., 1998), while stochastic volatility models allow separate state and observation noise (Jacquier et al., 1994; Kim et al., 1998). Our approach extends SV by allowing the variance transition to be a flexible, data-driven function of policy and carbon-market states, while keeping the model stable and interpretable—a key concern in moderate-sample energy-economics applications.

Gap and contribution. Existing work does not directly model the *joint* and potentially nonlinear role of CPU and carbon-price conditions in shaping clean energy equity volatility and tail risk. This paper provides that link with a methodology designed to be both flexible (able to learn interactions) and transparent (able to summarize state dependence in economically meaningful objects).

6 Data

6.1 Geographic focus and frequency

Europe, weekly frequency. We focus on Europe to align the outcome variable with the EU ETS policy environment. Weekly frequency matches the GCPU series, reduces microstructure noise, and keeps the sample size manageable (roughly ~ 600 observations). Daily robustness checks are reported in the appendix.

6.2 Clean energy equities (outcome)

- **Primary:** A European clean energy equity index or ETF (e.g., an S&P European Clean Energy index via standard terminals, or a European clean energy ETF). The specific series will be chosen based on data access and documented with a clear rationale.
- **Robustness:** ICLN (global) (BlackRock iShares, 2026); TAN (solar) (Invesco, 2026); S&P Global Clean Energy Transition Index (S&P Dow Jones Indices, 2026).
- **Placebo:** STOXX 600 Financials or STOXX 600 Industrials as relatively less climate-transition-exposed sectors where CPU effects should be weaker.

Log returns: $y_t = 100 \times (\log P_t - \log P_{t-1})$, using total return series where available.

6.3 Climate policy uncertainty

- **Baseline (weekly):** GCPU ([Figshare, 2024](#)).
- **Robustness (monthly):** CPU and decomposed $\text{CPU}^+/\text{CPU}^-$ ([PolicyUncertainty.com, 2026a,b](#)). Monthly series are aligned to weekly using (i) a step function (baseline), (ii) cubic spline interpolation, and (iii) MIDAS-style weighting (robustness).
- **Falsification:** General EPU ([Baker et al., 2016](#)). If climate-specific uncertainty matters, CPU measures should outperform EPU in predicting clean energy volatility.

6.4 EU ETS carbon prices

We use a front-month continuous EUA futures series (weekly close) from ICE ([Intercontinental Exchange \(ICE\), 2026](#)), cross-checked against Sandbag ([Sandbag / Carbon Market Watch, 2026](#)). Carbon covariates:

$$c_t = \Delta \log(\text{EUA}_t), \quad \ell_t = \log(\text{EUA}_t).$$

6.5 Financial conditions and energy price controls

- Credit spreads: ICE BofA US HY OAS ([Federal Reserve Bank of St. Louis \(FRED\), 2026](#)); [European iTraxx Crossover](#) if accessible.
- Rates: Euro area 2Y/10Y yields and the term spread.
- Equity volatility: VSTOXX to separate CPU-specific effects from broad market stress.
- Energy prices: TTF natural gas returns and Brent crude oil returns, given their role in EUA dynamics via fuel switching ([Creti et al., 2012](#)).

6.6 Sample

Main sample: 2013–present (EU ETS Phase 3 onward), approximately 600 weekly observations. An extended sample from 2008 is used for robustness. We use a consistent Wednesday-to-Wednesday weekly calendar.

7 Econometric Framework

7.1 Baseline: linear stochastic volatility with covariates

$$y_t = \exp(h_t/2) \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1), \quad (1)$$

$$h_t = \mu + \phi(h_{t-1} - \mu) + \beta' x_{t-1} + \sigma_\eta \eta_t, \quad \eta_t \sim \mathcal{N}(0, 1), \quad (2)$$

where x_{t-1} includes CPU/GCPU, EUA variables, and financial conditions. The coefficient vector β captures the average (linear) effect of the covariates on log-variance. This model is both a natural benchmark and the nested linear special case of our proposed specification.

7.2 Proposed: nonlinear stochastic volatility with observable inputs

Design choice: observable returns replace latent standardized shocks. A natural candidate for the nonlinear transition input would be the standardized shock $\varepsilon_{t-1} = y_{t-1} \exp(-h_{t-1}/2)$, but ε_{t-1} is *not observed*. It depends on h_{t-1} , which is latent. In a Bayesian MCMC framework the latent path $h_{1:T}$ is sampled and ε_{t-1} is available at each iteration as a deterministic transform; however, feeding a latent-dependent quantity into a flexible nonlinear function creates two practical concerns: (i) the highly nonlinear coupling between h_{t-1} and $g_\theta(\cdot)$ can produce difficult posterior geometries and slow MCMC mixing; (ii) identification of the NN weights becomes entangled with the posterior over h_{t-1} , because the same return y_t can be rationalized by different (h_{t-1}, g_θ) combinations.

We therefore use **observable lagged returns** y_{t-1} directly. This is clean, fully identified conditional on data, and creates no additional latent coupling. The network can learn how the *scale* and *sign* of returns interact with the policy-carbon state to drive future volatility. Formally:

$$\boxed{h_t = \mu + \phi(h_{t-1} - \mu) + g_\theta(h_{t-1}, y_{t-1}^+, y_{t-1}^-, x_{t-1}) + \sigma_\eta \eta_t}, \quad (3)$$

where:

- $y_{t-1}^+ = \max(y_{t-1}, 0)$ and $y_{t-1}^- = \min(y_{t-1}, 0)$ are the positive and negative parts of the lagged return, entered separately so the network can learn leverage-type sign asymmetries without requiring pre-specification;
- x_{t-1} is the vector of exogenous covariates (CPU, EUA log-return c_{t-1} , EUA log-level ℓ_{t-1} , credit spreads, VSTOXX, energy prices);
- $g_\theta : \mathbb{R}^d \rightarrow \mathbb{R}$ is a small feedforward neural network defined below.

Why raw returns are sufficient. While standardized shocks ε_{t-1} have unit variance by construction, the raw return y_{t-1} carries information about both the shock direction and its scale in volatility units. The network receives h_{t-1} as a separate input, so it can *implicitly* learn the scaling: a large negative return at a time of low h_{t-1} represents a very large standardized shock, and the network can approximate this relationship. Formally, since $y_{t-1} = \exp(h_{t-1}/2) \varepsilon_{t-1}$, the pair (y_{t-1}, h_{t-1}) is informationally equivalent to $(\varepsilon_{t-1}, h_{t-1})$ via an invertible (and monotone) mapping. Any function $f(\varepsilon_{t-1}, h_{t-1})$ can be expressed as $\tilde{f}(y_{t-1}, h_{t-1})$. The NN universal-approximation property ensures that g_θ can learn $\tilde{f}$ to arbitrary precision given sufficient capacity.

Architecture and stability.

- **Input vector:** $z_{t-1} = [h_{t-1}, y_{t-1}^+, y_{t-1}^-, x_{t-1}] \in \mathbb{R}^d$, with d typically 8–12.
- **Hidden layers:** One hidden layer with K units ($K \in \{8, 16\}$):

$$a = \tanh(W_1 z_{t-1} + b_1), \quad u_\theta = w_2' a + b_2,$$

where $W_1 \in \mathbb{R}^{K \times d}$, $b_1 \in \mathbb{R}^K$, $w_2 \in \mathbb{R}^K$, $b_2 \in \mathbb{R}$.

- **Bounded output:** $g_\theta(z_{t-1}) = s \cdot \tanh(u_\theta)$, so $|g_\theta| \leq s$ for a learnable or fixed scale $s > 0$ (e.g., $s = 0.5$).
- **Persistence constraint:** $\phi = \tanh(\tilde{\phi}) \in (-1, 1)$.
- **Nesting:** When $W_1 = 0$ and $b_1 = 0$, the network output reduces to $g_\theta = s \cdot \tanh(b_2)$, a constant, and the model collapses to a standard linear SV. With a single linear layer (identity activation) and one input, it nests the linear covariate SV exactly.

Heavy-tailed extension. The observation equation can accommodate excess kurtosis by replacing the Gaussian with a Student- t distribution:

$$y_t = \exp(h_t/2) \varepsilon_t, \quad \varepsilon_t \sim t_\nu, \quad (4)$$

where ν is estimated with a prior $\nu \sim \text{Gamma}(2, 0.1)$ (shifted to ensure $\nu > 2$).

8 Bayesian Estimation in Stan

We estimate the full model using Hamiltonian Monte Carlo (HMC) via the No-U-Turn Sampler (NUTS; Hoffman and Gelman, 2014) as implemented in Stan (Stan Development Team, 2024). Stan’s automatic differentiation provides exact gradients of the log-posterior with respect to all parameters—including the neural-network weights—which makes HMC efficient even in the high-dimensional joint space of latent volatilities and model parameters. This section provides a detailed description of the estimation procedure.

8.1 Joint posterior

The target is the joint posterior over the latent log-variance path $h_{1:T}$ and all model parameters Θ :

$$p(\Theta, h_{1:T} \mid y_{1:T}, X) \propto \underbrace{\prod_{t=1}^T p(y_t \mid h_t)}_{\text{observation likelihood}} \cdot \underbrace{p(h_1 \mid \Theta) \prod_{t=2}^T p(h_t \mid h_{t-1}, y_{t-1}, x_{t-1}, \Theta)}_{\text{state transition}} \cdot \underbrace{p(\Theta)}_{\text{priors}}, \quad (5)$$

where $\Theta = \{\mu, \tilde{\phi}, \sigma_\eta, W_1, b_1, w_2, b_2, s, (\nu)\}$ collects all parameters. Note that y_{t-1} and x_{t-1} in the transition density are *observed data*, not latent quantities, which keeps the posterior geometry cleaner than it would be if the transition depended on the latent standardized shock.

8.2 Likelihood components

Observation density. For the Gaussian case:

$$y_t \mid h_t \sim \mathcal{N}(0, \exp(h_t)), \quad \log p(y_t \mid h_t) = -\frac{1}{2} [\log(2\pi) + h_t + y_t^2 \exp(-h_t)]. \quad (6)$$

For the Student- t extension:

$$y_t \mid h_t, \nu \sim t_\nu(0, \exp(h_t)), \quad \log p(y_t \mid h_t, \nu) = \log \Gamma\left(\frac{\nu+1}{2}\right) - \log \Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2} \log(\nu\pi) - \frac{h_t}{2} - \frac{\nu+1}{2} \log\left(1 + \frac{y_t^2 \exp(-h_t)}{\nu}\right), \quad (7)$$

State transition. The transition density for $t \geq 2$ is:

$$h_t \mid h_{t-1}, y_{t-1}, x_{t-1}, \Theta \sim \mathcal{N}(\mu + \phi(h_{t-1} - \mu) + g_\theta(h_{t-1}, y_{t-1}^+, y_{t-1}^-, x_{t-1}), \sigma_\eta^2). \quad (8)$$

The initial state receives a stationary prior:

$$h_1 \sim \mathcal{N}\left(\mu, \frac{\sigma_\eta^2}{1 - \phi^2}\right). \quad (9)$$

8.3 Prior distributions

Table 1 summarizes the prior structure. All priors are weakly informative, designed to regularize the posterior without imposing strong beliefs.

Table 1: Prior distributions for all model parameters.

Parameter	Prior	Rationale
μ (log-var mean)	$\mathcal{N}(0, 5)$	Weakly informative; log-variance centered near zero
$\tilde{\phi}$ (raw persistence)	$\mathcal{N}(2, 1)$	Implies $\phi = \tanh(\tilde{\phi}) \approx 0.96$ a priori
σ_η (vol-of-vol)	Half- $\mathcal{N}(0, 1)$	Positive; prevents explosive paths
ν (degrees of freedom)	$2 + \text{Gamma}(2, 0.1)$	Ensures $\nu > 2$; mild prior on tail heaviness
$[W_1]_{ij}$ (NN layer 1)	$\mathcal{N}(0, \tau_w^2)$	Bayesian weight decay; $\tau_w \sim \text{Half-}\mathcal{N}(0, 1)$
$[b_1]_j$ (NN bias 1)	$\mathcal{N}(0, 1)$	Weakly informative
$[w_2]_j$ (NN layer 2)	$\mathcal{N}(0, \tau_w^2)$	Same hierarchical scale as layer 1
b_2 (NN bias 2)	$\mathcal{N}(0, 1)$	Weakly informative
s (output bound)	Half- $\mathcal{N}(0, 1)$	Controls amplitude of nonlinear term
τ_w (weight scale)	Half- $\mathcal{N}(0, 1)$	Hierarchical regularization of NN weights

Role of the hierarchical weight prior. The shared scale parameter τ_w on the NN weights plays the role of *Bayesian regularization*. When the data do not support nonlinearity, the posterior for τ_w concentrates near zero, effectively shrinking the network toward zero output and recovering the linear SV model. This is the Bayesian analogue of L2 weight decay and provides automatic complexity control without requiring a separate model-selection step (Piironen and Vehtari, 2017).

8.4 Non-centered parameterization for the latent path

Sampling $h_{1:T}$ directly (“centered parameterization”) can lead to strong posterior correlations between the latent states and (ϕ, σ_η) , producing slow mixing. We use a **non-centered parameterization** (Papaspiliopoulos et al., 2007). Define auxiliary i.i.d. standard normals

$\eta_1^{\text{raw}}, \dots, \eta_T^{\text{raw}} \sim \mathcal{N}(0, 1)$ and reconstruct the latent path deterministically:

$$h_1 = \mu + \frac{\sigma_\eta}{\sqrt{1 - \phi^2}} \eta_1^{\text{raw}}, \quad (10)$$

$$h_t = \mu + \phi(h_{t-1} - \mu) + g_\theta(h_{t-1}, y_{t-1}^+, y_{t-1}^-, x_{t-1}) + \sigma_\eta \eta_t^{\text{raw}}, \quad t = 2, \dots, T. \quad (11)$$

Stan samples $\eta_{1:T}^{\text{raw}}$ (unit normal), and $h_{1:T}$ is computed in the **transformed parameters** block. This decouples the “shape” of the path from σ_η and typically improves the effective sample size per iteration substantially.

8.5 Neural-network forward pass inside Stan

Stan does not have built-in neural-network layers, but the forward pass is straightforward to implement using matrix operations. For a single hidden layer with K units and tanh activation:

$$a_t = \tanh(W_1 z_{t-1} + b_1) \in \mathbb{R}^K, \quad (12)$$

$$g_\theta(z_{t-1}) = s \cdot \tanh(w'_2 a_t + b_2) \in \mathbb{R}, \quad (13)$$

where $z_{t-1} = (h_{t-1}, y_{t-1}^+, y_{t-1}^-, x'_{t-1})' \in \mathbb{R}^d$. Stan’s automatic differentiation computes $\nabla_\Theta \log p$ through this computation graph exactly, enabling NUTS to traverse the joint $(\eta_{1:T}^{\text{raw}}, \Theta)$ space efficiently.

8.6 Stan program structure

Below we provide the complete Stan program for the proposed model. The code implements a one-hidden-layer feedforward network with K units, bounded output, non-centered latent path, and optional Student- t observations.

Listing 1: Stan program for the nonlinear SV model with observable return inputs.

```

1 // nonlinear_sv.stan
2 // Nonlinear stochastic volatility with a small neural network transition
3 // Inputs: observable lagged returns (y+, y-) and exogenous covariates
4 // Estimation: NUTS (Hamiltonian Monte Carlo)
5
6 data {
7   int<lower=2> T;           // number of time periods
8   int<lower=1> D;           // dimension of exogenous covariates x
9   int<lower=1> K;           // number of hidden units
10  vector[T] y;              // observed log returns (percent)
11  matrix[T-1, D] X;         // exogenous covariates x_{t-1}, rows t=2..T
12  int<lower=0, upper=1> use_student_t; // 0 = Gaussian, 1 = Student-t
13 }
14
15 transformed data {
16   // Pre-compute positive and negative return parts
17   vector[T-1] y_pos;       // y_{t-1}^+ for t = 2..T
18   vector[T-1] y_neg;       // y_{t-1}^- for t = 2..T
19   int D_nn = D + 3;        // NN input dim: h_{t-1}, y+, y-, x_{t-1}
20
21   for (i in 1:(T-1)) {

```

```

22     y_pos[i] = fmax(y[i], 0.0);
23     y_neg[i] = fmin(y[i], 0.0);
24 }
25 }
26
27 parameters {
28     // --- SV parameters ---
29     real mu;                                // log-variance mean
30     real phi_raw;                           // raw persistence (unconstrained)
31     real<lower=0> sigma_eta;                 // volatility of log-variance
32
33     // --- NN parameters (one hidden layer) ---
34     matrix[K, D_nn] W1;                     // input-to-hidden weights
35     vector[K] b1;                           // hidden biases
36     vector[K] w2;                           // hidden-to-output weights
37     real b2;                                // output bias
38     real<lower=0> s;                         // output bound: |g| <= s
39
40     // --- Hierarchical weight scale ---
41     real<lower=0> tau_w;                     // shared scale for NN weights
42
43     // --- Latent path (non-centered) ---
44     vector[T] eta_raw;                      // auxiliary standard normals
45
46     // --- Heavy tails (optional) ---
47     real<lower=0> nu_minus2[use_student_t ? 1 : 0];
48 }
49
50 transformed parameters {
51     real phi = tanh(phi_raw);                // persistence in (-1, 1)
52     vector[T] h;                            // latent log-variance path
53
54     // Reconstruct h_1 (stationary distribution)
55     h[1] = mu + (sigma_eta / sqrt(1 - phi^2)) * eta_raw[1];
56
57     // Reconstruct h_2, ..., h_T via nonlinear transition
58     for (t in 2:T) {
59         // Assemble NN input vector: [h_{t-1}, y_{t-1}^+, y_{t-1}^-, x_{t-1}]
60         vector[D_nn] z;
61         z[1] = h[t-1];
62         z[2] = y_pos[t-1];
63         z[3] = y_neg[t-1];
64         z[4:D_nn] = X[t-1]';
65
66         // Forward pass: one hidden layer + bounded output
67         vector[K] a = tanh(W1 * z + b1);
68         real nn_out = s * tanh(dot_product(w2, a) + b2);
69
70         // Transition equation (non-centered)
71         h[t] = mu + phi * (h[t-1] - mu) + nn_out + sigma_eta * eta_raw[t];
72     }
73 }
74
75 model {
76     // === PRIORS ===
77     // SV parameters
78     mu ~ normal(0, 5);
79     phi_raw ~ normal(2, 1);                // prior centers phi near 0.96
80     sigma_eta ~ normal(0, 1);              // half-normal (declared <lower=0>)
81 }

```

```

82 // NN weights: hierarchical regularization
83 tau_w ~ normal(0, 1); // half-normal
84 to_vector(W1) ~ normal(0, tau_w);
85 b1 ~ normal(0, 1);
86 w2 ~ normal(0, tau_w);
87 b2 ~ normal(0, 1);
88 s ~ normal(0, 1); // half-normal
89
90 // Latent auxiliary variables
91 eta_raw ~ std_normal();
92
93 // Heavy tails (if enabled)
94 if (use_student_t) {
95     nu_minus2 ~ gamma(2, 0.1); // nu = nu_minus2 + 2 > 2
96 }
97
98 // === LIKELIHOOD ===
99 if (use_student_t) {
100     real nu = nu_minus2[1] + 2;
101     for (t in 1:T) {
102         y[t] ~ student_t(nu, 0, exp(h[t] / 2));
103     }
104 } else {
105     for (t in 1:T) {
106         y[t] ~ normal(0, exp(h[t] / 2));
107     }
108 }
109 }
110
111 generated quantities {
112     // Posterior predictive log-likelihood (for L00-CV)
113     vector[T] log_lik;
114
115     // One-step-ahead volatility forecast (h_{T+1} | data)
116     real h_forecast;
117     real vol_forecast;
118
119     // Log-likelihood computation
120     if (use_student_t) {
121         real nu = nu_minus2[1] + 2;
122         for (t in 1:T)
123             log_lik[t] = student_t_lpdf(y[t] | nu, 0, exp(h[t] / 2));
124     } else {
125         for (t in 1:T)
126             log_lik[t] = normal_lpdf(y[t] | 0, exp(h[t] / 2));
127     }
128
129     // Forecast: draw h_{T+1}
130     {
131         vector[D_nn] z_T;
132         z_T[1] = h[T];
133         z_T[2] = fmax(y[T], 0.0);
134         z_T[3] = fmin(y[T], 0.0);
135         // Note: x_T must be passed separately for true forecasting;
136         // here we use the last available row for illustration
137         // In practice, supply x_T as additional data
138         for (d in 1:D) z_T[3+d] = X[T-1, d]; // placeholder
139
140         vector[K] a_T = tanh(W1 * z_T + b1);
141         real nn_T = s * tanh(dot_product(w2, a_T) + b2);

```

```

142     h_forecast = mu + phi * (h[T] - mu) + nn_T
143               + sigma_eta * normal_rng(0, 1);
144     vol_forecast = exp(h_forecast / 2);
145 }
146 }

```

8.7 Running the sampler: practical details

Step 1: Data preparation (in R or Python). All covariates in x_{t-1} are **standardized** (zero mean, unit variance) before passing to Stan. This ensures the NN weights operate on a comparable scale and improves HMC geometry. Returns y_t are in percentage points (e.g., $y_t \in [-10, 10]$ typically for weekly data). The data list passed to Stan is:

```

stan_data = list(T = T, D = D, K = K,
                 y = y, X = X_lagged,
                 use_student_t = 1)

```

Step 2: Initialization. Good starting values accelerate warmup. We initialize from a pre-estimated linear SV model:

- $\mu^{(0)}, \tilde{\phi}^{(0)}, \sigma_\eta^{(0)}$: from a simple $\log\text{-}\chi^2$ approximation or a quick GARCH fit.
- NN weights: initialized near zero ($W_1^{(0)} \sim \mathcal{N}(0, 0.01)$), so the model starts close to the linear SV.
- $\eta_{1:T}^{\text{raw}(0)} = 0$ (flat initialization).

Step 3: Sampler configuration.

- **Chains:** 4 parallel chains.
- **Warmup:** 1,500 iterations per chain (with adaptation of step size and mass matrix).
- **Sampling:** 2,000 post-warmup draws per chain (8,000 total).
- **Tree depth:** `max_treedepth = 12` (increased from default 10 to accommodate the nonlinear geometry).
- **Target acceptance:** `adapt_delta = 0.90` (or 0.95 if divergences appear).

Step 4: Convergence diagnostics.

- **$\hat{R}$ statistic** (Gelman and Rubin, 1992): Require $\hat{R} < 1.01$ for all parameters and key quantiles of h_t .
- **Effective sample size (ESS):** Require bulk-ESS and tail-ESS > 400 for all monitored quantities.
- **Divergent transitions:** Monitor for divergences; if present, increase `adapt_delta` or reparameterize.

- **Energy diagnostics:** Check the Bayesian fraction of missing information (E-BFMI > 0.3).
- **Visual checks:** Trace plots, rank histograms, and pairs plots for key parameters (ϕ , σ_η , τ_w , s).

Step 5: Post-processing and inference. After sampling, posterior draws provide a complete probabilistic characterization of all quantities of interest:

- **Posterior summaries:** Means, medians, and 90% credible intervals for all parameters.
- **Filtered volatility:** The posterior mean of $\exp(h_t/2)$ provides the estimated volatility path; the 5th–95th percentile band provides uncertainty.
- **One-step-ahead predictive density:** For each posterior draw $(\Theta^{(s)}, h_T^{(s)})$, simulate $h_{T+1}^{(s)}$ from the transition and then $y_{T+1}^{(s)} \sim \mathcal{N}(0, \exp(h_{T+1}^{(s)}))$. The empirical distribution of $y_{T+1}^{(s)}$ is the posterior predictive density.
- **VaR and ES:** Computed directly from quantiles of the predictive draws.
- **LOO-CV:** The `log_lik` output enables Pareto-smoothed importance sampling leave-one-out cross-validation (Vehtari et al., 2017) via the `loo` R package.

8.8 Out-of-sample forecasting protocol

We use an **expanding-window** design to generate a sequence of genuine out-of-sample forecasts:

1. **Initial training window:** 2013–2018 (≈ 300 weeks). Estimate the model in Stan.
2. **Validation window:** 2019 (≈ 52 weeks). Used for architecture selection ($K \in \{8, 16\}$, Gaussian vs. t_ν).
3. **Test window:** 2020–present ($\approx 250+$ weeks). Generate one-step-ahead forecasts.
4. **Re-estimation frequency:** Every 13 weeks (quarterly), the model is re-estimated on all data up to the current point. Between re-estimations, we update the latent state using the most recent posterior draws but do not re-run MCMC (a “warm-start” approach).
5. **Forecast construction:** At each test-period date t , we use the posterior draws from the most recent estimation to simulate $h_{t+1}^{(s)}$ and compute the one-step-ahead predictive density, VaR, and ES.

8.9 Computational requirements

With $T \approx 600$, $K = 16$ hidden units, and $D \approx 8$ covariates, the total number of sampled parameters is approximately:

$$\underbrace{T}_{\eta^{\text{raw}}} + \underbrace{3}_{\mu, \phi, \sigma_\eta} + \underbrace{K(D+3) + K + K + 1 + 1 + 1}_{W_1, b_1, w_2, b_2, s, \tau_w} + \underbrace{1}_{\nu} \approx 600 + 3 + 16 \times 11 + 16 + 16 + 3 + 1 = 815.$$

This is well within Stan’s capabilities. On a modern workstation (8-core CPU), a single chain with 3,500 iterations (1,500 warmup + 2,000 sampling) should complete in 1–4 hours, depending on tree depth. Four chains can run in parallel. The full out-of-sample exercise (re-estimating quarterly over ~ 5 years ≈ 20 re-estimations) requires approximately 80–320 CPU-hours total, feasible on a standard academic computing cluster.

8.10 Model comparison via LOO-CV and posterior predictive checks

LOO-CV. We use PSIS-LOO (Vehtari et al., 2017) to compare the nonlinear SV, linear SV, and the restricted model ($g_\theta \equiv 0$). The `loo_compare()` function in R provides formal comparisons. If the Pareto $\hat{k}$ diagnostics indicate problematic observations ($\hat{k} > 0.7$), we use moment-matching or refit those folds.

Posterior predictive checks. We verify model adequacy by simulating from the posterior predictive distribution and comparing the implied distribution of squared returns, absolute returns, and autocorrelation of squared returns against the observed data.

9 Identification and Causal Language

This is a *predictive and mechanism-discovery* study. We avoid causal language and interpret results as conditional risk relationships rather than causal effects. All covariates enter lagged (x_{t-1}), and we include broad controls (VSTOXX, credit spreads, energy prices) to separate CPU-specific dynamics from generic “risk-off” episodes.

Identification support.

- **Placebo test:** CPU should have weaker predictive power for volatility in non-climate-exposed European sectors (e.g., STOXX 600 Financials). A near-null result supports the clean-energy channel.
- **CPU⁺ vs. CPU⁻:** If the channel is policy-specific, tightening- vs. loosening-related uncertainty may have different volatility implications.
- **CPU vs. EPU:** Climate-specific uncertainty should outperform general EPU in forecasting clean energy volatility, helping to rule out purely generic uncertainty confounds.
- **Event windows (extension):** Focused analyses around major EU climate-policy milestones (ETS reforms, Fit for 55 votes) with interactions by CPU/EUA state.

10 Evaluation

10.1 Benchmarks

1. GARCH(1,1) with covariates (Bollerslev, 1986)
2. GJR-GARCH(1,1) with covariates (Glosten et al., 1993)

3. EGARCH(1,1) with covariates (Nelson, 1991)
4. Linear SV with covariates (Eq. 2), estimated in Stan
5. Regime-switching SV (So et al., 1998)
6. HAR-RV (Corsi, 2009) (if intraday data permits)

All variance equations include the same covariates x_{t-1} to ensure apples-to-apples comparisons. The linear SV benchmark is estimated in the same Stan framework (with $g_\theta \equiv 0$) to ensure differences reflect the nonlinear component rather than estimation methodology.

10.2 Statistical evaluation

- **Density forecasts:** Out-of-sample log predictive score (LPS), reported overall and conditional on CPU regimes (high/low). The Bayesian predictive density is computed by averaging over posterior draws.
- **Volatility loss:** QLIKE loss (Patton, 2011) using squared returns as the proxy.
- **Tail risk:** VaR backtests at 1% and 5% (unconditional and conditional coverage (Christoffersen, 1998); dynamic quantile test (Engle and Manganelli, 2004)). ES backtests via joint scoring functions (Fissler and Ziegel, 2016).
- **Formal comparison:** Diebold–Mariano tests (Diebold and Mariano, 1995) and Model Confidence Set (Hansen et al., 2011).
- **Bayesian comparison:** LOO-CV via PSIS-LOO (Vehtari et al., 2017) and stacking weights.

10.3 Economic significance: volatility-timing portfolio

To translate forecasting gains into economic terms, we implement a simple volatility-targeting strategy. The investor allocates between clean energy equities and a risk-free asset to maintain a target annualized volatility (e.g., 15%) using each model’s one-step-ahead forecast. We report realized Sharpe ratios, maximum drawdowns, and certainty-equivalent returns. This makes it easier to interpret whether better forecasts matter in basis-point terms.

10.4 Mechanism and interpretability

1. **State-dependent impulse responses:** Fix conditioning states at quantile combinations (high/low CPU $\times$ falling/rising EUA $\times$ tight/loose financial conditions). For each posterior draw, simulate the average volatility path following a one-standard-deviation negative return shock, comparing peaks and half-lives across regimes. Report posterior means and 90% credible bands of the IRFs.
2. **Marginal effect surfaces:** Compute $\partial g_\theta / \partial \text{CPU}$ numerically (finite differences) over a (CPU quantile $\times$ EUA quantile) grid, for each posterior draw. Visualize the posterior mean as a heatmap.

3. **Amplification ratio:** $\mathcal{A} = \sum_{j=1}^H \Delta h_{t+j}^{\text{high CPU}} / \sum_{j=1}^H \Delta h_{t+j}^{\text{low CPU}}$ —the ratio of cumulative IRFs across CPU regimes. Report the full posterior distribution of $\mathcal{A}$ across carbon-price states to quantify the interaction with uncertainty.
4. **Ablation analysis:** Re-estimate the model with input blocks removed (CPU, EUA, financial conditions, return sign) one at a time and record the LOO-CV score change. This clarifies which channels carry the most predictive content.

11 Policy Implications

This section previews the policy-relevant messages we expect the empirical results to sharpen.

The financial stability value of credible carbon pricing. If CPU-driven volatility amplification is strongest when EUA prices are falling, then credibility and price stability in the ETS matter not only for emissions incentives but also for financial risk. Put differently: predictable carbon pricing may reduce the probability of “risk cascades” in transition assets. This speaks directly to the design and calibration of stability mechanisms (e.g., the MSR) and to policy communication.

Cost of policy reversals. The amplification ratio $\mathcal{A}$ provides a concrete way to talk about the hidden cost of policy uncertainty: how much worse are volatility dynamics under high CPU relative to low CPU? If $\mathcal{A}$ is large (e.g., above 2), it quantifies how policy ambiguity can magnify downside risk in a way that standard linear models would miss. The full Bayesian posterior over $\mathcal{A}$ also conveys how certain we are about this magnification.

Risk management for transition portfolios. Institutional investors increasingly hold clean energy and climate-aligned exposures. If the nonlinear model materially improves tail-risk forecasts, it implies that commonly used linear VaR/volatility tools may understate downside risk during high-CPU episodes, potentially leading to unintended concentration risk precisely in stressed regimes.

Carbon market design. If major ETS reforms (such as the MSR’s introduction in 2019) coincide with weaker state-dependent amplification, that would be consistent with a financial-risk benefit of stabilizing the carbon price signal. This is a policy-relevant complement to the usual efficiency and emissions-focused evaluations of carbon-market design.

12 Robustness Checks

1. **Alternative equity proxies:** Global (ICLN) vs. European; solar (TAN) vs. broad clean energy; individual large-cap clean energy firms.
2. **Placebo:** Non-climate European sectors (financials, industrials). A weak/null effect supports channel specificity.

3. **Alternative carbon series:** ICE vs. EEX vs. Sandbag; alternative rolling rules.
4. **Alternative uncertainty:** CPU vs. GCPU; $\text{CPU}^+/\text{CPU}^-$; EPU as falsification.
5. **Frequency:** Daily estimation (appendix); monthly aggregation.
6. **Subsamples:** Pre/post MSR (2019); pre/post Fit for 55 (2021); COVID sensitivity.
7. **Controls:** With/without gas and oil returns; with/without VSTOXX.
8. **Model diagnostics:** Prior sensitivity analysis (doubling/halving prior scales); alternative NN sizes ($K \in \{4, 8, 16, 32\}$); centered vs. non-centered parameterization comparison.
9. **Observable vs. latent shocks:** As a robustness check, estimate a variant that feeds the MCMC-sampled $\hat{\varepsilon}_{t-1} = y_{t-1} \exp(-h_{t-1}/2)$ into g_θ and compare posterior summaries and out-of-sample scores.

13 Expected Contributions

1. **Substantive (energy economics):** Evidence on how CPU relates to clean energy equity *volatility dynamics*, with effects that vary systematically with carbon-market conditions.
2. **Policy:** Quantification of how policy uncertainty and carbon pricing interact to generate nonlinear financial risk, informing discussions on credibility and ETS stability tools.
3. **Practical:** Improved tail-risk forecasting for transition-sensitive portfolios, illustrated via a volatility-timing strategy.
4. **Methodological:** A parsimonious, stability-constrained nonlinear SV model estimated via full Bayesian inference in Stan, paired with interpretable outputs (IRFs, marginal effects, ablations) and explicit nesting of linear benchmarks.

14 Feasibility and Timeline

14.1 Feasibility

The required series are publicly available or accessible via standard data terminals: CPU/GCPU ([PolicyUncertainty.com](#), 2026a; [Figshare](#), 2024); EUA prices ([Sandbag / Carbon Market Watch](#), 2026; [Intercontinental Exchange \(ICE\)](#), 2026); clean energy equities ([BlackRock iShares](#), 2026; [Invesco](#), 2026; [S&P Dow Jones Indices](#), 2026); financial conditions ([Federal Reserve Bank of St. Louis \(FRED\)](#), 2026). Estimation uses Stan ([Stan Development Team](#), 2024), which is freely available and well-supported via CmdStan, RStan, or PyStan. With approximately 815 parameters (including the latent path), NUTS sampling is feasible on a standard multi-core workstation.

14.2 Timeline (14 weeks)

- **Weeks 1–2:** Data acquisition, cleaning, alignment. Descriptive analysis by CPU regime.
- **Weeks 3–4:** Benchmark estimation (GARCH variants in R; linear SV in Stan). Build the evaluation pipeline.
- **Week 5:** Simulation study on synthetic data: generate data from a known nonlinear SV, estimate in Stan, verify parameter recovery and coverage.
- **Weeks 6–8:** Nonlinear SV estimation in Stan; architecture selection (K , Gaussian vs. t_ν); convergence tuning (NCP, `adapt_delta`, `treedepth`).
- **Weeks 9–10:** Out-of-sample evaluation, tail-risk backtests, portfolio exercise, LOO-CV comparisons.
- **Weeks 11–12:** IRFs, marginal effects, robustness checks, placebo tests, ablation analysis.
- **Weeks 13–14:** Write-up and replication package (Stan code, R/Python scripts, data documentation).

14.3 Reproducibility

We will include a data appendix documenting each series, transformations, and construction rules. The Stan model file, R/Python driver scripts, and all post-processing code will be shared as a seeded, deterministic repository. A single replication script will reproduce all tables and figures from raw data to final output.

15 Risks and Mitigation

- **Overfitting.** We mitigate this with a small network ($K \leq 16$), bounded outputs, a hierarchical weight prior (τ_w), and strict out-of-sample evaluation via LOO-CV and expanding-window forecasts. If the nonlinear model does not outperform benchmarks out of sample, we treat that as an informative result: linear dynamics are sufficient here.
- **MCMC mixing.** The non-centered parameterization, standardized inputs, and careful initialization should produce adequate mixing. If divergences persist, we increase `adapt_delta`, reduce K , or switch to a centered parameterization for a subset of the path. We report all convergence diagnostics transparently.
- **Frequency mismatch.** Weekly GCPU is the baseline; monthly CPU is aligned using three methods and sensitivity is reported.
- **Sample size (~600 weekly).** The architecture is chosen specifically for this data regime ($K \in \{8, 16\}$). The hierarchical prior provides additional shrinkage. Learning curves (LOO-CV vs. training size) and posterior predictive checks are reported.
- **Confounding.** We include financial conditions controls, VSTOXX, and energy prices, and we run placebo tests and CPU vs. EPU comparisons.

- **Reviewer concern about “black box.”** We emphasize interpretability (IRFs, marginal effects, ablations), formal nesting of linear SV, the transparent Bayesian framework with full posterior uncertainty, and a clear motivation: the state space is high-dimensional and the functional form is not obvious ex ante. A flexible but constrained approach is less ad hoc than manually specifying many interactions and thresholds.
- **EU ETS structural breaks.** We run subsample analyses around key reforms (MSR in 2019; Fit for 55 starting in 2021). The nonlinear transition can also adapt to shifts more smoothly than hard regime breaks.
- **Stan run time.** If estimation is slower than expected, we can reduce K , thin the posterior draws, or use CmdStan’s multi-threading (`reduce_sum`) to parallelize the likelihood evaluation within chains.

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